

Vaults

Arched roofs of stone, or vaults, have been a fundamental element of architecture since early times. To create large open spaces, roofs had to span great distances, and for many early builders vaults were the only solution. In its simplest form a vault is an arch, and the simplest of all is the barrel vault (also called the waggon or tunnel vault), which is semicircular in section, like a cylinder cut in half lengthwise. Early examples of barrel vaulting go all the way back to the Sumerians, ancient Egyptians and, especially, the Romans. The Romans also used groin vaults, in which two barrel vaults cross at right angles, as at the transept of a church. (The groins are the arcs traced by the intersecting planes.) Romanesque developments enabled the joining of vaults of different spans and heights; but the great advance was the Gothic rib vault, in which the arcs evolved into raised ribs and load-bearing arches. Some rib vaulting became decorative and highly complex – such as sexpartite (six-panelled) vaults and fan vaulting – in the great Gothic churches and cathedrals built across medieval Europe.

